

Productive Failure Report

What the naive single-signal memory baseline actually does on the GTM workload — measured, not asserted.

System Conversational Memory Intelligence System

Author Dheeraj Pranav

Baseline Approach D — store-all, single-signal retrieval

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Protocol ../experiments/baseline_protocol.md

Data baseline_results.csv · error_examples.jsonl

HEADLINE

The naive baseline passes **0 of 11** queries (overall accuracy **0.0**). It is not a weak *retriever* — it is fast (p95 **0.05 ms**) and stays far under budget (**92** of 2000 tokens injected). It fails because whole classes of requirement live **outside** what a similarity signal can express. A superseded fact beats the current one **80%** of the time; a foreign tenant's memory leaked into top-k on **7 of 11** queries; a health-PII record was injected **3** times, once on a query with no relevant memory at all. Every failure is **structural**: re-ordering the same candidate set does not add a validity field, a tenant boundary, or a PII gate. **HYPOTHESIS SUPPORTED**

1. Hypothesis (pre-registered)

From `baseline_protocol.md §1`: the strongest of the simpler designs — single-signal retrieval over stored chunks — fails on the GTM workload **not because its retriever is weak, but because temporal validity, resolved meaning, admission control, and tenant isolation are orthogonal to any similarity signal**. If naive retrieval instead handled the adversarial conditions, Deliverable 1's design would be over-built and should be cut. The baseline was constructed to give that outcome a fair chance (§7).

2. Baseline design

Approach D from `reconstruction/failure_analysis.md`, implemented literally: **store every chunk → retrieve top-k by cosine similarity → inject in similarity order until a token budget is hit**. Similarity is lexical TF-IDF cosine (pure stdlib, deterministic, no model download — justification and threats in the protocol). Deliberately absent, and *that absence is the experiment*: admission control, typed/resolved facts, temporal validity, multi-signal ranking, index-level tenant partitioning, and any abstention threshold. Fixed parameters: `k=5`, `budget=2000`, `seed=20260722`.

3. Workload

A fixed, seeded, adversarial-by-construction GTM dataset: **44 memories** (1 PII, 3 foreign-tenant, 68% filler/duplicate), **two tenants that both have an “Acme” account** with near-identical notes, and **11 queries** covering all six required conditions. Traps are hand-authored so a single similarity signal is *forced* to expose each failure — stale facts carry *more* query-term overlap than their corrections, so the lexical bias runs in the baseline's favour, making this a harder test to pass, not an easier one.

REQUIRED CONDITION	QUERIES	PROBES
Irrelevant + contradictory retrievable	q01, q02, q03, q11	F4, F8
Preferences that change (supersession)	q01, q02, q03	F4
Long conversation, constrained budget	q08	F3, F8
Multiple tenants, similar wording	q05, q06	F4, F10
Sensitive info that must not surface	q07 (+ leaks into q04, q09)	F11
Cold start / no relevant memory	q09, q10	cold-start

4. Results

METRIC	MEASURED	READING
overall_accuracy	0.000 (0/11)	no query passes every condition-appropriate check
recall_at_k (answerable)	0.333 (3/9)	the current fact is usually not even in the top-5
supersession_failure_rate	0.800 (4/5)	the stale fact outranks the current one four times in five
inversion_failures (F7)	1	the inverting utterance beats the resolved fact by 30 ranks
cross_tenant_leak_rate (F10-probed)	0.500 (1/2)	— but see the emergent rate below
cross-tenant leak (all queries)	7 / 11	foreign-tenant memory in top-k far beyond the designed cases
pii_exposure_count (F11)	3	one PII record injected on 3 different queries, incl. a cold-start
coldstart_abstention_failure_rate	1.000 (2/2)	never abstains; injects noise when nothing is relevant
wasted_topk_slot_rate	0.556 (25/45)	over half of retrieved slots are filler/duplicate
duplicate_or_filler_ratio (corpus)	0.682	the store is mostly noise; nothing gates admission
avg / max injected tokens	92.1 / 105	the 2000-token budget <i>never bound</i> — size is not the problem
latency p50 / p95	≈0.04 / 0.05 ms	the baseline is trivially cheap; its failures are not a cost problem

The two rows that matter most are the ones that look benign: **budget never bound** and **latency is negligible**. They rule out the easy story (“just optimise retrieval / add more budget”). At k=5 the failure is *what* is selected and *whether it should be stored at all* — not how much or how fast.

4.1 Four failures, concretely (from `error_examples.jsonl`)

Q02 — “WHO IS THE DECISION MAKER AT ACME?” **F4 + F10**

Rank 1: `m003` stale — “Ravi Menon, VP Engineering” (0.752). Rank 2: `m102` **from tenant T2 (Contoso)** — “Dana Whitfield” (0.717). The *correct* current fact (Priya owns it now, `m004`) is at **rank 31**. A departed contact and a *different customer’s* decision-maker both beat the truth.

Q04 — “DOES ACME NEED SSO?” **F7 + F11**

Rank 1: `m005` — “I don’t think we need SSO for the pilot” (0.463). The resolved fact (“put it in — non-negotiable for production”, `m006`) is at **rank 31**. The injected context asserts the *inverse* of the account’s real

position — and rank 3 is a **health-PII record** (`m009` : a colleague’s chemotherapy leave and personal mobile) on a question about SSO.

Q09 — “RENEWAL DATE FOR THE GLOBEX ACCOUNT?” `COLD-START + F10 + F11`

Nothing on Globex exists; the baseline should abstain. Instead it injects `m103` (T2’s **confidential 85k budget**), an unrelated SSO utterance, and the PII record `m009`. The worst possible answer to “I have nothing on file” is another tenant’s deal size plus someone’s medical data.

Q01 — “WHAT CRM DOES ACME USE?” `F4 + F8`

The entire top-3 is **duplicate recitations of the stale Salesforce fact** (`d001/d002/d003` , all 0.44). The current fact (migrated to HubSpot, `m002`) is at rank 14. Storing everything let one obsolete fact manufacture its own majority.

5. Failure taxonomy (measured against D1 §6)

#	FAILURE (D1)	MEASURED	RECOVERABLE?
F4	Superseded fact served as current	<code>0.80</code>	yes
F7	Chunk inverts meaning out of context	<code>1 CASE, GOLD @31</code>	yes
F8	Index saturated with low-value duplicates	<code>0.556 SLOTS</code>	yes
F10	Cross-tenant memory returned	<code>7/11 QUERIES</code>	no
F11	Sensitive data retained & surfaced	<code>3 EXPOSURES</code>	no
cold	No abstention on empty retrieval	<code>2/2</code>	yes

The baseline reproduced **2 of D1’s 5 unrecoverable failures** (F10, F11), and both fired hard. That is the whole argument for the architecture in one line: a system can tolerate retrieving a mediocre memory; it cannot tolerate retrieving *another tenant’s* memory or a record it should never have stored.

WHAT THIS RUN DOES NOT SHOW — STATED TO KEEP THE RESULT HONEST

F1, F2 (correction across sessions; quadratic prompt growth) need a multi-turn write path; **F5, F6** (summarisation loss / drift) need a summariser; **F9** (deleted-yet-retrievable) needs a delete-then-requery test. A retrieval-only baseline structurally cannot exercise these — they move to the Deliverable 6 verification harness. **F3 did not reproduce here**: the buried EU-residency dealbreaker (`m020`) happened to surface at rank 1 because the query shared its terms, so q08 failed on F8/F10, not on a buried-fact miss. Reported as a *non-confirmation*, not glossed over. The dataset is small and author-built: it shows the *existence and character* of each failure, not population frequencies.

6. Proposed response — and the D1 tags it retires

Each measured failure maps to a mechanism already decided in Deliverable 2, so the baseline does more than confirm the problem — it prices the fix:

FAILURE	MEASURED	MECHANISM (D2 DECISION)
F4	0.80 supersession	Bi-temporal validity + invalidation on the write path (ADR-001) — ranker only ever orders currently-valid facts
F7	gold buried at rank 31	Extraction into typed, <i>resolved</i> claims at admission — store the fact, not the utterance
F8	0.556 wasted slots	Admission salience + dedup; consolidation that cites, not repeats
F10	7/11 leaked	Index-level tenant partitioning as an invariant below the retrieval API (not a post-filter)
F11	3 exposures	Presidio deterministic PII gate at admission + deletion backstop (ADR-003)
cold-start	2/2 no-abstain	Similarity-floor abstention: return “nothing on file” below threshold

Deliverable 1 **TO MEASURE** tags now retired to numbers: the `failure_analysis.md` D-3 claim (“proportion of retrieved slots occupied by non-informative or duplicate content”) is now **0.556**; the D-1/B-3 claim that a single similarity signal cannot prefer current over stale is now **0.80** supersession-failure. The tags that remain open (restatement-token share, summariser drift) require the mechanisms this baseline deliberately lacks and stay **TO MEASURE** for Deliverable 6.

7. Falsification

What would have refuted the hypothesis: high recall@k with failures concentrated in low-similarity *misses* — the kind a better encoder fixes. That is not what happened.

- **The gaps are validity/identity gaps, not relevance gaps.** The correct fact sat at rank **31** in two separate cases while a stale or foreign chunk sat at rank 1–2. No re-ordering *within* a similarity score closes a 30-rank gap that encodes *which fact is true* or *whose tenant it is* — those axes are not in the score.
- **A stronger embedder makes F10 worse, not better.** The cross-tenant leaks come from near-identical notes across two “Acme” accounts; higher semantic fidelity raises their similarity, so a better model leaks *more* confidently. Isolation cannot be earned by accuracy.
- **Cost and budget are ruled out.** p95 latency 0.05 ms and 92/2000 tokens injected mean the failures are not artefacts of a starved budget or a slow index.

Honest caveats. One targeted failure (F3 buried-fact) did not reproduce on this dataset, and the abstention result is measured against a reference threshold the baseline does not apply. Neither rescues the naive design: the two *unrecoverable* failures it did produce (F10, F11) are sufficient on their own to reject store-everything single-signal retrieval for a multi-tenant, regulated GTM setting. The hypothesis stands, and the Deliverable 4 design must own each mechanism in §6 — each is now backed by a measured failure rate, not an assertion.

Reproducibility. Pure Python standard library, fixed seed (20260722), no network or API key. `cd experiments/naive_baseline && python3 build_dataset.py && python3 evaluate.py` reproduces every *failure* metric here byte-for-byte; only the two wall-clock latency figures vary run-to-run (in the 3rd decimal). Method, metric definitions, and threats to validity: `experiments/baseline_protocol.md`.

AI assistance disclosure. Claude helped implement the harness and draft this report. The hypothesis, the adversarial dataset design, the failure→mechanism mapping, and the falsification argument are mine and are what I will defend in review. Every number is machine-produced by `evaluate.py` from the committed dataset; none is hand-entered.